SAIMAI LAU

Robotics, Controls & Human-Centered Systems

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EDUCATION

Ph.D. Mechanical Engineering

University of Wisconsin–Madison

Jun 2026 — Present | Madison, WI

- Osteo Engineering Lab — Advisor: **Prof. David Dean**
- Robotics, Controls & Sensing track; **Bendy Bot** project

M.S. Mechanical Engineering

Stanford University

Sept 2024 — Jun 2026 | Stanford, CA

- Depth: **Robotics & Kinematics**
- Breadth: **Biomechanical Engineering**; **Automatic Controls**
- Faculty Advisor: **Prof. Allison Okamura** Research Advisor: **Prof. Michaëlle Mayalu**
- GPA: 3.91

B.S. Mechanical Engineering (Controls & Robotics)
B.S. Mathematics (Applied Science)

University of California San Diego

Aug 2021 — Jun 2024 | La Jolla, CA

- GPA: 3.95; **Provost Honors every quarter**; completed both degrees in **three years**
- Magna Cum Laude

RESEARCH EXPERIENCE

Graduate Research Assistant

Osteo Engineering Lab (UW–Madison ME) — Advisor: Prof. David Dean

Jun 2026 — Present | Madison

- Designing controls for the **Bendy Bot**, a robotic system for patient-specific bone fixation plate bending.
- Collaboration with the HAMMER advanced-manufacturing research community.

Graduate Researcher

ILIAD Lab (Stanford CS) — with J. Grannen; Advisor: Prof. Dorsa Sadigh

Mar 2026 — Jun 2026 | Stanford

- **Co-author**: "Overcoming Distribution Shifts with Autonomous Embodied Data Collection," *IEEE ICRA 2026 Workshop (Beyond Teleop)*.
- Built an autonomous robot data-collection pipeline (TidyBot++ base + Franka FR3 arm + iPhone RGBD) using ROS2 and multimodal LLMs to target distribution gaps for model fine-tuning.

Graduate Researcher

Mayalu Lab (Stanford ME) — Advisor: Prof. Michaëlle Mayalu

Sep 2024 — Jun 2026 | Stanford

- **Co-author**: "Optimal Control Strategies for Cutaneous Inflammatory Disease Treatment Using a Predictive Multi-scale Model," presented at *ACC 2026*.
- Designed and implemented a **Pontryagin-based hybrid optimal-control** framework; reformulated propositions and proofs for rigor.
- Built a data-driven gut-skin microbial ODE model (MICOM flux-balance analysis, Hill-function fitting) as a plant model for optimal treatment design.

Undergraduate Researcher

Multi-Agent Robotics (MURO) Lab — Advisor: Prof. Sonia Martínez

Jun 2023 — Jun 2024 | UC San Diego

- Implemented **distributed MEKF-SLAM** and **cyclic-pursuit** algorithms on TurtleBots (ROS2; LiDAR, IMU, RGB-D camera object detection, Ultrasonic beacons).
- Developed ROS2 utilities for distributed SLAM integration and sensor-fault recovery.
- Co-presented: "Distributed Variational Inference for Online Supervised Learning," *Southern California Robotics Symposium 2023*.

Undergraduate Researcher

Boomerang Aerodynamics Research — Advisor: Prof. Prasad Gudem

Jan 2023 — Jun 2024 | UC San Diego

- Derived analytical expressions for lift, drag, and torque in tribladder designs; parametric CAD in SOLIDWORKS.
- Field validation via **UWB** and **drone-based 3D trajectory tracking**; compared prediction vs experiment.

PUBLICATIONS

- Mo, P., **Lau, S.**, Gao, J., Sadigh, D., Grannen, J. (2026). *Overcoming Distribution Shifts with Autonomous Embodied Data Collection*. IEEE ICRA 2026 Workshop (Beyond Teleop).
- Subedar, G., **Lau, S.**, Mayalu, M.N. (2026). *Optimal Control Strategies for Cutaneous Inflammatory Disease Treatment Using a Predictive Multi-scale Model*. American Control Conference (ACC).

PRESENTATIONS

- Paritosh, P., **Lau, S.**, Atanasov, N., Martínez, S. (2023). *Distributed Variational Inference for Online Supervised Learning*. Southern California Robotics Symposium 2023, UC Irvine.

AWARDS & HONORS

UCSD Provost Honors (2021–2024)
HK IVE Youth EV Design — 3rd (2020)
HK PolyU Robotic Challenge — 2nd (2018)
ARL Korea Championship Build Award — Best Structural Design (2017)

SELECTED PROJECTS

Voice-Controlled Bimanual Pick-and-Place Robot

Stanford ME326

2026

- Systems integrator for a bimanual mobile manipulator team project: built grasp execution server, dropoff service, 3-phase PID navigator, **AnyGrasp**/**SAM3** integration, GCP voice pipeline (STT/TTS), and an LLM-based autonomous commander.
- Developed keyboard teleop GUI, demo recorder, and debugging tools; tested in **MuJoCo** simulation with custom ArUco scenes and deployed on hardware.

Haptic Game Interface

Stanford ME327

2025

- Designed and built an electromechanical device with **real-time force feedback** for a modified Block Breaker game.
- Implemented position control, collision rendering, and dynamic stiffness modulation on embedded hardware.

Autonomous RC Car

UCSD MAE 148

2024

- Embedded Linux vehicle with LiDAR, GNSS, RGB-D camera, and on-edge YOLO for tennis ball retrieval.

On-Demand Ropeless Fishing Gear Tracker

UCSD MAE 156A–B (Capstone)

2024

- Waterproof submerged tracking system (∼1000 ft) with Android app for sustainable fisheries.

TECHNICAL & PROGRAMMING

Java | Python | MATLAB | Julia | C++ | ROS2 | Bash | Git | LaTeX | HTML/CSS/JS | MySQL | PHP

Control & Estimation
PID; EKF/UKF/MEKF; LQR; MPC; Pontryagin Minimum Principle; Forward–Backward Sweep; Optimal & Learning-Based Control; Spectral Methods

ML & Perception
PyTorch; TensorFlow; OpenCV; YOLO; Bayesian Inference; Gaussian Processes; POMDP; Markov Processes

Modeling & Simulation
MuJoCo; Gazebo; RViz; Simulink; SOLIDWORKS; FEA (Abaqus, Ansys); Cardiovascular Modeling (SimVascular); Musculoskeletal modeling (OpenSim)

Hardware & Embedded
Arduino; Jetson Nano; Raspberry Pi; VESC; LiDAR; IMU; UWB; RGB-D; GNSS-R; 3D printing; laser cutting; soldering; CNC machining

LANGUAGES

English (Fluent); Cantonese (Native); Mandarin (Native)